- Ass 4 due Fri 30 Oct @ 4pm
- Ass 4 Q3: "I haven't learnt Lagrange multipliers..."

For Binomial, p_1 & p_2 subject to $p_1 + p_2 = 1$ vs p_1, $1-p_1$

Recap:

(1) We can summarize a one-way ANOVA via the table:

Source	df	SS	MS	F	P(F
Groups	$J-1$	$\sum_j n_j(\bar{Y}_{j\cdot} - \bar{Y}_{\cdot\cdot})^2$	$SS_{Groups}/J-1$	MS_{Group}/MS_{Resid}	↑
Resids	$N-J$	$\sum_j \sum_i (Y_{ji} - \bar{Y}_{j\cdot})^2$	$SS_{Resids}/N-J$		
Total	$N-1$				

$P(F_{J-1,N-J} \geq F)$

(2) A contrast is any linear combination of the group means $\sum_{j=1}^{J} a_j \mu_j$ such that $\sum_{j=1}^{J} a_j = 0$

(3) The one-way ANOVA layout can be parametrised via the "Sum constraint":

$$Y_{ji} = \mu + \alpha_j + \varepsilon_{ji}, \quad \text{with} \sum_{j=1}^{J} \alpha_j = 0$$

Then, $\mu = \frac{\mu_1 + \mu_2 + \ldots + \mu_J}{J}$ is the average mean across all groups

& each $\alpha_j = \mu_j - \mu$ is a contrast that compares every group to the average

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"Contrast parametrisation/ constraint": The one-way ANOVA layout can be rewritten as

$$Y_{ji} = \mu + \alpha_j + \varepsilon_{ji}$$

where $\alpha_1 \equiv 0$ for identifiability.

Then, μ is interpreted as the mean of group 1 ($\mu \equiv \mu_1$)

& each α_j represents the difference between group j & group 1:

$$\alpha_j = \mu_j - \mu_1 = \mu_j - \mu \quad , \quad j = 2, 3, \dots, J$$

When is the "contrast parametrisation" useful?

- If one of the groups is a baseline/reference/placebo, & every other group is compared to this reference group.

In R (Matlab, Python, etc...) the default is to use the contrast parametrisation, with "group 1" set alpha-numerically.

Remember, all three parametrisations specify the exact same model:

Parametrisation	Analogy
"mean" $\mu_1, \mu_2, \dots, \mu_J$	I am 171 cm tall, my brother is 173 cm tall & my sister is 165 cm tall
"sum" $\mu, \alpha_1, \alpha_2, \dots, \alpha_J$ with $\sum \alpha_j = 0$	The average height of my family is 169.67 cm. I am 1.33 cm above average, & my brother is 3.33 cm above average.
"contrast" $\mu_1 \equiv \mu, \alpha_2, \alpha_3, \dots, \alpha_J$ note $\alpha_1 \equiv 0$	I am 171 cm tall. My brother is 2 cm taller than me. My sister is 6 cm shorter than me.

Example 8.7 (tooth growth vs Vitamin C)

RDEMO

Two-way ANOVA

If the groups in our study are obtained by crossing two-or-more factors (each at two or more levels), then we should respect the study design & consider a two-way ANOVA model instead:

The most general two-way setup is given by

$$Y_{jki} = \mu + \alpha_j + \beta_k + \gamma_{jk} + \varepsilon_{jki},$$

where $j = 1, 2, \ldots, J$ indexes level of factor A
$k = 1, 2, \ldots, K$ indexes level of factor B
& $i = 1, 2, \ldots, n_{jk}$ indexes individuals

& $\varepsilon_{jki} \sim N(0, \sigma^2)$

parameters: $1 + J + K + JK$

groups: JK

To make this model identifiable, we have to impose constraints:

(1) "Sum parametrization" with

$$\sum_{j=1}^{J} \alpha_j = 0, \quad \sum_{k=1}^{K} \beta_k = 0, \quad \sum_{j=1}^{J} \gamma_{jk} = 0 = \sum_{k=1}^{K} \gamma_{jk}$$

$\Leftrightarrow$ $\alpha_\cdot = 0$, $\beta_\cdot = 0$, $\gamma_{\cdot k} = 0$ for $k = 1, 2, \ldots, K$
$\gamma_{j\cdot} = 0$ for $j = 1, 2, \ldots, J$
[$\gamma_{\cdot\cdot} = 0$ automatically]

(2) "contrast parametrisation" with

$\alpha_1 \equiv 0$, $\beta_1 \equiv 0$, $\gamma_{1k} = 0 \quad k = 1, 2, \ldots, K$

$\gamma_{j1} = 0 \quad j = 1, 2, \ldots, J$

[$\gamma_{11} = 0$ automatically]

Default = R / Matlab / Python..

In either parametrisation, the parameters can generically interpreted as

α_j's are the "main effects" / "marginal effects" of factor A

β_k's " " " " " " of factor B

γ_{jk}'s are the interaction effects between factor A & B

If we focus on the contrast parametrisation, then

μ = mean of the "first" group (factor A set to level 1 & factor B set to level 1)

α_j = the expected change when changing factor A to level j (from level 1) [holding factor B at level 1]

β_k = " " " " factor B to level k (from level 1) [holding factor A at level 1]

γ_{jk} = the additional change in the mean when changing BOTH factor A to level j AND factor B to level k simultaneously.

When we have a two-way ANOVA, the first point of interest is to test for interactions. If there are interactions, then it may NOT be meaningful to discuss main effects.

Only if there are NO interactions, then we can move on to talk about effects.

Example 8.7 (tooth growth, continued).

Our 6 treatment groups can actually be considered as a 2×3 design instead:

$$(OJ, VC) \times (\text{Low}, \text{Medium}, \text{High})$$

Supp type — Dosages.

R DEMO

Note: we used "lm" to fit ANOVA (!) models in R. This is because ANOVA is a special case of a linear model.

§ 9 Linear models

In general, a linear model assumes that the data has the following structure:

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$$

where $\underset{\sim}{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$ is a (long) vector of responses, one row for each individual response

$$X = \begin{pmatrix} X_{11} & X_{12} & X_{13} & \cdots & X_{1p} \\ X_{21} & X_{22} & X_{23} & \cdots & X_{2p} \\ \vdots & & & & \\ X_{n1} & X_{n2} & X_{n3} & \cdots & X_{np} \end{pmatrix}$$

sample size ↓ n, # predictors ↓ p

is an $n \times p$ matrix of explanatory variables / predictors ("design matrix")

$\underset{\sim}{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}$ is a $p \times 1$ vector of regression parameters ("slopes")

& $\underset{\sim}{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$ is a vector of random deviations away from the "line".

Example 9.1 (One-way ANOVA as a linear model)

If we use the default contrast parameterization, then

$$Y_{ji} = \mu + \alpha_j + \varepsilon_{ji}, \text{ with } \alpha_1 \equiv 0$$

$$\underset{\sim}{Y} = \begin{pmatrix} Y_{11} \\ Y_{12} \\ Y_{13} \\ \vdots \\ Y_{1n_1} \\ Y_{21} \\ Y_{22} \\ Y_{23} \\ \vdots \\ Y_{2n_2} \\ Y_{31} \\ Y_{32} \\ \vdots \\ Y_{3n_3} \\ \vdots \\ Y_{J1} \\ Y_{J2} \\ \vdots \\ Y_{Jn_J} \end{pmatrix}, \quad \underset{\sim}{X} = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \\ 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ \vdots & & & & \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 1 & 0 \cdots & 0 \\ 1 & 0 & 1 & 0 \cdots & 0 \\ \vdots & & & & \\ 1 & 0 & 1 & 0 \cdots & 0 \\ \vdots & & & & \\ 1 & 0 & \cdots & & 0 \\ 1 & 0 & \cdots & & 0 \\ \vdots & & & & \\ 1 & 0 & \cdots & & 0 \end{pmatrix}, \quad \underset{\sim}{\beta} = \begin{pmatrix} \mu \\ \alpha_2 \\ \alpha_3 \\ \vdots \\ \alpha_J \end{pmatrix}$$

So we have shown that one-way ANOVA is a linear model.

Example 9.2

In a shotput study, distance was measured along the armspan. A model was proposed:

$$\text{Distance}_i = \beta_0 + \beta_1 \text{armspan}_i + \beta_2 \text{armspan}_i^2 + \varepsilon_i$$

Q: Is this a linear model? Debate over lunch